DAVINA WHATA

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PROFESSIONAL SUMMARY

Geographic Information Systems (GIS) and Geospatial professional with 3 years of experience supporting utility infrastructure mapping, spatial data management, ecological research, and academic instruction across telecommunications, higher education, and research environments. Skilled in GIS data integration, spatial analysis, network connectivity validation, and geospatial quality assurance, with a strong ability to manage complex spatial datasets, identify data inconsistencies, and produce accurate mapping solutions. Experienced working in collaborative research and technical environments, supporting fiber-optic network design, wildlife data analysis, and geospatial education through effective data management, technical problem-solving, and spatial analysis.

TECHNICAL SKILLS

GIS Platforms: ArcGIS Pro, ArcGIS Online, ArcMap, QGIS, VETRO FiberMap, Fiber Manager, Field Maps, Survey123

Spatial Data Engineering: Geodatabases, GIS data integration, utility records, feature/attribute editing, topology QA/QC, network connectivity, raster/vector processing, spatial statistics

Python & Data Tools: Python, ArcPy, GeoPandas, pandas, GDAL, Jupyter Notebook, GPS/GNSS, CAD-to-GIS, KML/KMZ, remote sensing

PROFESSIONAL EXPERIENCE

THAYER POWER & COMMUNICATION **Pataskala, Ohio**

GIS Technician Intern. *May – Aug 2026*

- Updated and validated fiberoptic network designs across 8 projects in 4 Indiana counties using ArcGIS Pro and VETRO FiberMap.
- Resolved fiber circuitry and connectivity errors by tracing customer drops through splitters and splice cases to feeder tie points.
- Corrected topology, slack-loop, handhole, and cable-routing errors, preparing network designs for engineering review.

MICHIGAN TECHNOLOGICAL UNIVERSITY **Houghton, Mi**

Graduate Research Assistant *Aug – Dec 2025*

- Validated 1,000+ GPS/GNSS-referenced wildlife records, correcting coordinate and attribute inconsistencies for ecological research.
- Applied spatial analysis techniques to examine 500+ GPS/GNSS-referenced wildlife records, supporting the identification of wildlife distribution patterns and interpretation of ecological research findings.
- Managed geospatial QA/QC for a 4-person research team, maintaining organized records and producing GIS maps.

SOL PLAATJE UNIVERSITY **Northern Cape, South Africa**

Academic Tutor *Feb 2022 – Nov 2024*

- Tutored 30 first-year undergraduate students in GIS Techniques and Environmental Archaeology, providing academic support in spatial analysis, mapping techniques, and geospatial concepts.
- Guided students in GIS data interpretation, spatial data integration, and practical mapping applications, strengthening their understanding of technical concepts.
- Provided individualized academic guidance, helping students develop analytical, technical, and problem-solving skills.

EDUCATION

MICHIGAN TECHNOLOGICAL UNIVERSITY **Houghton, Mi**

Master of Science in Geographic Information Science December 2026

- Relevant Coursework: Data Science for Environmental Sensing, Python for Modern GIS, Applied spatial statistics, Advanced GIS concepts & analysis, Remote sensing of the environment

SOL PLAATJE UNIVERSITY **Northern Cape, South Africa**

Bachelor of Social Sciences Honours Degree, Environmental Science. May 2025

- Thesis title: An investigation of ancient climatic variations during the Holocene period through rock art analysis and lichenometry: A case study of Wildebeest Kuil Rock Art Centre.

PROJECTS

Utility Data Integration & Validation — Network and municipal assets

Integrated engineering and field records into GIS features; checked attributes, topology, and network consistency across utility datasets.

Spatial Method Evaluation — Geostatistics and remote sensing

Compared IDW, Natural Neighbor, and kriging using cross-validation; analyzed Landsat imagery and spatial patterns with Moran's I and kernel density.

Python Geospatial Processing — ArcPy, GeoPandas, and GDAL

Used Python GIS libraries for data extraction, proximity analysis, raster resampling, and processing workflows; inspected outputs for accuracy.

ADDITIONAL INFORMATION

Languages: Shona (native), English (fluent), Afrikaans (basic), Zulu (basic)

Interests and Volunteer Activities: Fundraising for menstrual health and hygiene initiatives, women's empowerment, and community outreach.